Binomial Model

Pricing

$$\begin{cases} q_1 + q_2 = 1, \\ q_1 S_0 u + q_2 S_0 d = S_0(1 + r_n), \end{cases}$$

Contingent claim at time 0:

$$\Phi(S_0) = \frac{1}{(1+r)^n} \sum_{k=0}^{n} \binom{n}{k} q_u^k q_d^{n-k} \phi^{(k)u;(n-k)d}$$

No-Arbitrage and Completeness

$d < 1 + r < u$

- NA $\Rightarrow$ market complete
- Every contingent claim is replicable
- Replication price = risk-neutral price

Replicating Portfolio (iff NA)

$$\begin{cases} \phi_{t+1}^u = x_t(1+r) + y_t S_t u = V_{t+1}^u, \\ \phi_{t+1}^d = x_t(1+r) + y_t S_t d = V_{t+1}^d \end{cases}$$

Arbitrage Strategy Construction – Arbitrage exists if $\exists h$ such that: $V_0^h = 0, \quad \mathbb{P}(V_T^h \geq 0) = 1, \quad \mathbb{P}(V_T^h > 0) > 0$

Cox-Ross-Rubinstein (CRR)

$1 + R = e^{r\Delta t}$

$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}} = 1/u$

$q_u = \frac{e^{r\Delta t} - d}{u - d}, \quad q_d = 1 - q_u$

Stochastic Calculus

Standard Brownian Motion (W_t) Properties

- Initial value: $W_0 = 0$
- Increments: $W_t - W_s \sim \mathcal{N}(0, t-s)$
- $\mathbb{E}[W_t] = 0$
- $\mathrm{Var}(W_t) = t$
- $\mathrm{Cov}(W_s, W_t) = \min(s, t)$
- Itō rules: $(dW_t)^2 = dt, \quad (dt)^2 = 0$

Geometric Brownian Motion (GBM)

SDE: $dX_t = \mu X_t dt + \sigma X_t dW_t$

Solution: $X_t = X_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$

Itō Lemma

$(Z_t)_{t \geq 0}$ defined as $Z_t = f(X_t, t)$ satisfies:

$$\begin{aligned} dZ_t &= \frac{\partial f}{\partial t}(X_t, t)\, dt + \frac{\partial f}{\partial x}(X_t, t)\, dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(X_t, t)\, \sigma^2(t)\, dt \\ &= \left(\frac{\partial f}{\partial t} + \mu \frac{\partial f}{\partial x} + \frac{\sigma^2}{2}\frac{\partial^2 f}{\partial x^2} \right)(X_t, t)\, dt + \sigma(t) \frac{\partial f}{\partial x}(X_t, t)\, dW_t \end{aligned}$$

The stoch. proc. $(Z_t)_{t \geq 0}$ defined as $Z_t = f(X_t^1, \ldots, X_t^n, t)$ satisfies:

$$dZ_t = \frac{\partial f}{\partial t} dt + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} dX_t^i + \frac{1}{2} \sum_{i,j=1}^{n} \sigma^i \sigma^j \rho_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} dt$$

One dimensional formula:
$dZ_t = \left(\mu f'(X_t) + \frac{\sigma^2}{2} f''(X_t) \right) dt + \sigma f'(X_t)\, dW_t$

Correlated d-dimensional Brownian Motion

- $B_t = (B_t^1, \ldots, B_t^n)$, $\mathbb{E}[B_t^i] = 0$
- $\mathrm{Cov}(B_t^i, B_t^j) = \rho_{ij} t$, $d\langle B^i, B^j \rangle_t = \rho_{ij} dt$
- Correlation construction (2D): $B_t^1 = \sqrt{1 - \rho^2} W_t^1 + \rho W_t^2$, $B_t^2 = W_t^2$ with $W^1 \perp W^2$.

Multidimensional Itō's Lemma: For $f(t, X_t^1, X_t^2)$ and $dX_t^i = \mu^i dt + \sigma^i dB_t^i$: $df = \left\{ \frac{\partial f}{\partial t} + \sum_i \mu^i f_{x_i} + \frac{1}{2} \sum_{i,j} \rho_{ij} \sigma^i \sigma^j f_{x_i x_j} \right\} dt + \sum_i f_{x_i} \sigma^i dB_t^i$

Martingale Condition

M_t is a martingale iff $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$, $0 \leq s < t$. In differential form, the DRIFT term must vanish. In particular, if $dZ_t = \Phi_t dW_t, \Phi \in \mathcal{M}^2$, then (Z_t) is a martingale.

Black & Scholes

Black-Scholes Setup

$$\begin{cases} dB_t = rB_t dt \\ dS_t = S_t(\alpha dt + \sigma dW_t) \\ S_t = S_0 \exp\{(\alpha - \frac{1}{2}\sigma^2)t + \sigma W_t\} \end{cases}$$

Black-Scholes PDE

$$\begin{cases} F_t + rSF_S + \frac{\sigma^2}{2} S^2 F_{SS} - rF = 0 \\ F(T, S) = \Phi(S_T) \end{cases}$$

Vanilla Call

$$C(S_t, t) = S_t N(d_1) - Ke^{-r(T-t)} N(d_2)$$

$$d_1 = \frac{\ln(S_t/K) + (r + \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}$$

Discrete Dividends Call

$$C(S_0, 0) = (S_0 - D)N(\hat{d}_1) - Ke^{-rT} N(\hat{d}_2)$$

$$\hat{d}_1 = \frac{\ln((S_0 - D)/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$\hat{d}_2 = \hat{d}_1 - \sigma\sqrt{T}$$

Continuous Dividends Call

$$C(S_t, t) = S_t e^{-\delta(T-t)} N(d_1') - Ke^{-r(T-t)} N(d_2')$$

$$d_1' = \frac{\ln(S_t/K) + (r - \delta + \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2' = d_1' - \sigma\sqrt{T-t}$$

Vanilla Put

$$P(S_t, t) = Ke^{-r(T-t)} N(-d_2) - S_t N(-d_1)$$

(Use same d_1, d_2 as Vanilla Call)

Discrete Dividends Put

$$P(S_0, 0) = Ke^{-rT} N(-\hat{d}_2) - (S_0 - D)N(-\hat{d}_1)$$

(Use same $\hat{d}_1, \hat{d}_2$ as Discrete Call)

Continuous Dividends Put

$$P(S_t, t) = Ke^{-r(T-t)} N(-d_2') - S_t e^{-\delta(T-t)} N(-d_1')$$

(Use same d_1', d_2' as Continuous Call)

Dividends Definitions

Continuous: $D = S_0(1 - e^{-\delta T})$

Discrete: $D = \sum_i d_i e^{-rT_i}$

Probabilities of ITM

$$\mathbb{P}(\text{Call ITM}) = \mathbb{P}(S_T > K) = 1 - \mathbb{P}\left(Z < \frac{\ln(K/S_0) - (\alpha - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \right)$$

$$\mathbb{Q}(\text{Call ITM}) = 1 - \mathbb{Q}\left(Z < \frac{\ln(K/S_0) - (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \right) = N(d_2)$$

European Call Greeks

$$\Delta_c = N(d_1), \quad \Gamma_c = \frac{\varphi(d_1)}{S\sigma\sqrt{T-t}}$$

$$\rho_c = K(T-t)e^{-r(T-t)} N(d_2)$$

$$\mathcal{V}_c = S\varphi(d_1)\sqrt{T-t}$$

$$\Theta_c = -\frac{S\sigma\varphi(d_1)}{2\sqrt{T-t}} - rKe^{-r(T-t)} N(d_2)$$

European Put Greeks

$$\Delta_p = N(d_1) - 1, \quad \Gamma_p = \Gamma_c$$

$$\rho_p = -K(T-t)e^{-r(T-t)} N(-d_2)$$

$$\mathcal{V}_p = \mathcal{V}_c$$

$$\Theta_p = -\frac{S\sigma\varphi(d_1)}{2\sqrt{T-t}} + rKe^{-r(T-t)} N(-d_2)$$

Normal Density: $\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$

Hedging & Others

Put-Call Parity

- No dividends: $C_t - P_t = S_t - Ke^{-r(T-t)}$
- Discrete dividends: $C_t - P_t = S_t - D - Ke^{-rT}$
- Continuous dividends: $C_t - P_t = S_t e^{-q(T-t)} - Ke^{-r(T-t)}$

Hedging Portfolio: $V = xS + yC_h$

Greeks for Stock:
$\Delta_S = 1, \quad \Gamma_S = 0, \quad \mathcal{V}_S = 0, \quad \rho_S = 0, \quad \Theta_S = 0.$

Option Spreads

- **Bull Spread**: buy $C(K_{\text{low}})$, sell $C(K_{\text{high}})$
- **Bear Spread**: sell P_1, buy $P_2(K_1 < K_2)$
- **Butterfly**: buy $C(K_1)$, sell $2C(K_2)$, buy $C(K_3)$ ($K_2 \approx$ avg)

Combinations

- **Straddle**: $P + C$, $K_P = K_C$
- **Strangle**: $P + C$, $K_P < K_C$
- **Strip**: $2P + C$, **Strap**: $P + 2C$

Inequalities for Options

- $C_t \leq S_t; \quad P_t \leq Ke^{-r(T-t)}$
- $C_t \geq S_t - Ke^{-r(T-t)}; \quad P_t \geq Ke^{-r(T-t)} - S_t$
- $S_t - K \leq C_t^A - P_t^A \leq S_t - Ke^{-r(T-t)}$

Meta Theorem

$R = \#$ BM, $N = \#$ Risky Assets.
No Arbitrage $\iff R \geq N$
Completeness $\iff R \leq N$
If market is incomplete ($R > N$), choose $\theta \in \mathbb{R}^R$ such that:
$\sigma \cdot \underline{\theta} = r \cdot \underline{1} - \alpha$
NA holds if σ has full rank (i.e. $R = N$ and σ is invertible).

Exotic Options

- **Cash-or-Nothing**: Pays K if $S_T > L$ (Call).
  Price $C = Ke^{-r(T-t)}N(d_2)$.
  Put ($S_T < L$): $P = Ke^{-r(T-t)}N(-d_2)$.
- **Asset-or-Nothing**: Pays $S_T 1_{S_T > K}$ (Call).
  Price $C = S_t N(d_1)$.
  Put ($S_T < K$): $P = S_t N(-d_1)$.
- **Chooser**: Right to choose at $T_1 < T_2$ if Call or Put.
  $\max(C_{T_1}, P_{T_1}) = C_{T_1} + \Phi(S_{T_1})$ where $\Phi(x) = [Ke^{-r(T_2-T_1)} - x]^+$
  $CH_0 = \mathbb{E}^{\mathbb{Q}}[e^{-rT_1}(C_{T_1} + \Phi(S_{T_1}))]$
  $CH_0 = C(S_0, K, T_2) + P(S_0, Ke^{-r(T_2-T_1)}, T_1)$
- **Asian**: $\left(\frac{1}{T+1}\sum S_{t_i} - K\right)^+$
- **Lookback**: Floating Strike ($S_T - \min S_u$), Fixed Strike ($\max S_u - K$).

Barrier Options: K strike, L barrier.
$a = (L/S_t)^{\frac{2r}{\sigma^2}-1}, b = (L/S_t)^{\frac{2r}{\sigma^2}+1}$.
$d_1 = \frac{1}{\sigma\sqrt{T-t}}\left[\ln\frac{S_t}{K} + (r+\sigma^2/2)(T-t)\right]$
$d_2 = \frac{1}{\sigma\sqrt{T-t}}\left[\ln\frac{S_t}{K} + (r-\sigma^2/2)(T-t)\right]$
$d_3 = \frac{1}{\sigma\sqrt{T-t}}\left[\ln\frac{S_t}{L} + (r+\sigma^2/2)(T-t)\right]$
$d_4 = \frac{1}{\sigma\sqrt{T-t}}\left[\ln\frac{S_t}{L} + (r-\sigma^2/2)(T-t)\right]$
$d_5 = \frac{1}{\sigma\sqrt{T-t}}\left[\ln\frac{S_t}{L} - (r-\sigma^2/2)(T-t)\right]$
$d_6 = \frac{1}{\sigma\sqrt{T-t}}\left[\ln\frac{S_t}{L} - (r+\sigma^2/2)(T-t)\right]$
$d_7 = \frac{1}{\sigma\sqrt{T-t}}\left[\ln\frac{S_tK}{L^2} - (r-\sigma^2/2)(T-t)\right]$
$d_8 = \frac{1}{\sigma\sqrt{T-t}}\left[\ln\frac{S_tK}{L^2} - (r+\sigma^2/2)(T-t)\right]$

In-Out Parity: $F_{IN} + F_{OUT} = F_{Vanilla}$

Down-and-In (DI) ($S_t > L$)

- **Call**:
  If $K > L$: $C_{DI} = (L/S_t)^{\frac{2r}{\sigma^2}-1}C(L^2/S_t, t)$
  If $K < L$: $C_{DI} = C(S_t, t) - C_{DO}$ (Use C_{DO} below)
- **Put**:
  If $K > L$: $P_{DI} = P(S_t, t) - P_{DO}$ (Use P_{DO} below)
  If $K < L$: $P_{DI} = (L/S_t)^{\frac{2r}{\sigma^2}-1}P(L^2/S_t, t)$

Down-and-Out (DO) ($S_t > L$)

- **Call**:
  If $K > L$: $C_{DO} = C(S_t, t) - C_{DI}$
  If $K < L$: $C_{DO} = S_t[N(d_3) - b(1 - N(d_6))] - Ke^{-r(T-t)}[N(d_4) - a(1 - N(d_5))]$
- **Put**:
  If $K > L$: $P_{DO} = Ke^{-r(T-t)}[N(d_4) - N(d_2) - a(N(d_7) - N(d_5))] - S_t[N(d_3) - N(d_1) - b(N(d_8) - N(d_6))]$
  If $K < L$: $P_{DO} = 0$ (Knocked out $S > L > K$ impossible for Put payoff)

Up-and-In (UI) ($S_t < L$)

- **Call**:
  If $K < L$: $C_{UI} = C(S_t, t) - C_{UO}$ (Use C_{UO} below)
  If $K > L$: $C_{UI} = (L/S_t)^{\frac{2r}{\sigma^2}-1}C(L^2/S_t, t)$
- **Put**:
  If $K < L$: $P_{UI} = (L/S_t)^{\frac{2r}{\sigma^2}-1}P(L^2/S_t, t)$
  If $K > L$: $P_{UI} = P(S_t, t) - P_{UO}$ (Use P_{UO} below)

Up-and-Out (UO) ($S_t < L$)

- **Call**:
  If $K < L$: $C_{UO} = S_t[N(d_1) - N(d_3) - b(N(d_6) - N(d_8))] - Ke^{-r(T-t)}[N(d_2) - N(d_4) - a(N(d_5) - N(d_7))]$
  If $K > L$: $C_{UO} = 0$
- **Put**:
  If $K < L$: $P_{UO} = P(S_t, t) - P_{UI}$
  If $K > L$: $P_{UO} = Ke^{-r(T-t)}[N(d_6) - b(1 - N(d_3))] - S_t[N(d_5) - a(1 - N(d_4))]$

Futures Options (Black 76) F_0 is the current future/forward price.
Call on Future/Forward: $C = e^{-rT}[F_0N(d_1) - KN(d_2)]$
Put on Future/Forward: $P = e^{-rT}[KN(-d_2) - F_0N(-d_1)]$
$d_1 = \frac{\ln(F_0/K) + \frac{1}{2}\sigma^2T}{\sigma\sqrt{T}}; \quad d_2 = d_1 - \sigma\sqrt{T}$

Pure Currency Options

Dynamics under Domestic $\mathbb{Q}$: $\tilde{S}_t^f = S_t^f \cdot X_t$
$d\tilde{S}_t^f = \tilde{S}_t^f[r_d dt + (\sigma_X + \sigma_f)dW_t^{\mathbb{Q}}]$
$C(t, \tilde{S}_t^f) = \tilde{S}_t^f N(d_1) - Ke^{-r_d(T-t)}N(d_2)$
$d_1 = \frac{\ln(\tilde{S}_t^f/K) + (r_d + \frac{1}{2}\|\sigma_X + \sigma_f\|^2)(T-t)}{\|\sigma_X + \sigma_f\|\sqrt{T-t}}$
Domestic Market Price of Risk: $\lambda_f = \lambda_d - \sigma_X$

Interest Rates

Affine Term Structure (ATS)
$p(t,T) = \exp\{A(t,T) - B(t,T)r_t\}$
Requires $\mu(r,t) = \alpha(t)r + \beta(t)$ and $\omega(r,t) = \sqrt{\gamma(t)r + \delta(t)}$
where $dr_t = \mu(r_t, t)dt + \omega(r_t, t)dW_t$.
System: $\begin{cases} B_t + \alpha(t)B - \frac{1}{2}\gamma(t)B^2 = -1; & B(T,T) = 0 \\ A_t - \beta(t)B + \frac{1}{2}\delta(t)B^2 = 0; & A(T,T) = 0 \end{cases}$
For a calibrated model: $p(t,T) = \exp\{-\int_t^T f^*(0,s)ds\}$
$Z^{HL}(t,T) = \exp(-A^{HL}(t,T) - B^{HL}(t,T)r_t)$ (Ho-Lee)
$Z^{HW}(t,T) = \exp(-A^{HW}(t,T) - B^{HW}(t,T)r_t)$ (Hull-White)

Vasicek Model $\alpha(t) = -a, \beta(t) = ab, \gamma(t) = 0, \delta(t) = \sigma^2$.
$dr_t = a(b - r_t)dt + \sigma dW_t$
$B(t,T) = \frac{1-e^{-a(T-t)}}{a}$
$A(t,T) = \frac{(B(t,T) - (T-t))(a^2b - \sigma^2/2)}{a^2} - \frac{\sigma^2B^2(t,T)}{4a}$
$r_t = r_0e^{-at} + b(1 - e^{-at}) + \sigma\int_0^t e^{-a(t-s)}dW_s$
$\mathbb{E}[r_t] = r_0e^{-at} + b(1 - e^{-at})$

Ho-Lee Model $\alpha(t) = 0, \beta(t) = \theta_t, \gamma(t) = 0, \delta(t) = \sigma^2$.
$dr_t = \theta_t dt + \sigma dW_t$
$B(t,T) = T - t$
$A(t,T) = \int_t^T \theta_s(s - T)ds + \frac{\sigma^2(T-t)^3}{6}$
Theta formula: $\theta_t^{HL} = \frac{\partial f(0,t)}{\partial t} + \sigma^2 t$

Hull-White Model $\alpha(t) = -a, \beta(t) = \theta_t, \gamma(t) = 0, \delta(t) = \sigma^2$.
$dr_t = (\theta_t - ar_t)dt + \sigma dW_t$
$B(t,T) = \frac{1-e^{-a(T-t)}}{a}$
$A(t,T) = \int_t^T [\frac{1}{2}\sigma^2B^2(s,T) - \theta_sB(s,T)]ds$
Theta formula: $\theta_t^{HW} = \frac{\partial f(0,t)}{\partial t} + af(0,t) + \frac{\sigma^2}{2a}(1 - e^{-2at})$

Option Pricing on Zero-Coupon Bonds
Use the normalized price $p(t,T) = \exp\{A(t,T) - B(t,T)r_t\}$,
same for the strike K: ALWAYS divide for N.
$C(t,T,K,T^*) = N\left[p(t,T^*)N(d_1) - \frac{K}{N}p(t,T)N(d_2)\right]$
$d_1 = \frac{1}{\sigma_p}\left[\ln\left(\frac{p(t,T^*)}{(K/N)p(t,T)}\right) + \frac{1}{2}\sigma_p^2\right], \quad d_2 = d_1 - \sigma_p$

Option Pricing (Vasicek)
$\sigma_p = \sigma\frac{1-e^{-a(T^*-T)}}{a}\sqrt{\frac{1-e^{-2aT}}{2a}}$

Option Pricing (Hull-White)
$\sigma_p = \sigma\frac{1-e^{-a(T^*-T)}}{a}\sqrt{\frac{1-e^{-2aT}}{2a}}$

Option Pricing (Ho-Lee)
$\sigma_p = \sigma(T^* - T)\sqrt{T}$

Libor & Swap

Variables: **Variables**:
$L(0, T_{i-1}, T_i)$: Forward LIBOR rate at $t = 0$.
$P(0, T_i)$: Zero-coupon bond price.
$\bar{\sigma}$: Forward Implied Volatility.
$N(\cdot)$: CDF.

Black-76 Formula for a Caplet
$\text{Cpl}(0) = M \cdot \alpha \cdot P(0, T_i)[L(0, T_{i-1}, T_i)N(d_1) - RN(d_2)]$
Auxiliary:
$d_1 = \frac{\ln(L/R) + \frac{1}{2}\Sigma_i^2(0, T_{i-1})}{\Sigma_i(0, T_{i-1})}$
$d_2 = d_1 - \Sigma_i$
Volatility Scaling: $\Sigma_i(0, T_{i-1}) = \bar{\sigma}\sqrt{T_{i-1}}$

Zero-Coupon Bond: $P(0,T) = \frac{1}{1 + L_{\text{spot}} \cdot T}$

Forward Rate: $L(0, T_{i-1}, T_i) = \frac{P(0, T_{i-1})}{P(0, T_i) - 1}$ scaled by $\frac{1}{\alpha}$

Swaps
$V_{\text{float}} = N\sum\tau_iP(0, T_i)L_i$
$V_{\text{fixed}} = NK\sum\tau_iP(0, T_i) = NK \cdot PVBP$
Par Swap Rate: $S(0) = \frac{\sum\ldots L_i}{\sum\ldots} = \frac{\text{Floating Leg}}{PVBP}$
Payer Swap Value: $N(S(0) - K)PVBP$
Receiver Swap Value: $N(K - S(0))PVBP$

Swaption Price (Black 76)
Payer Swaption $= N \cdot PVBP[SN(d_1) - KN(d_2)]$
Receiver Swaption $= N \cdot PVBP[KN(-d_2) - SN(-d_1)]$
$d_1 = \frac{\ln(S/K) + \frac{1}{2}\sigma_S^2T}{\sigma_S\sqrt{T}}$